Prodigy Anesthesia Quick Review

Vascular Surgery

What are the most common symptoms seen in patients who are presenting for carotid endarterectomy?	Patients often present with symptoms such as unilateral weakness or numbness, unilateral blindness, a recent history of transient ischemic attacks or cerebrovascular accident. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 406.*
What are the most common indications for carotid endarterectomy (CEA)?	CEA is indicated for patients who exhibit symptomatic carotid artery occlusion with a narrowing of 60-90%. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 406.*
Under which types of anesthesia can carotid endarterectomy be performed?	CEA can be perform with the patient under general anesthesia or under local anesthesia with the patient awake. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 406.*
What is the mortality rate of postoperative stroke versus postoperative myocardial infarction following carotid endarterectomy?	The mortality rate in patients who suffer a stroke following carotid endarterectomy is 15% versus 49% for those who suffer an MI following a carotid endarterectomy. *Nagelhout JJ, Zaglaniczny KL. Nurse Anesthesia. 4th ed. Philadelphia, PA: WB Saunders Company; 2010: 547.*
Carotid surgery can be performed under regional anesthesia by blocking the nerves originating from which spinal levels?	Carotid surgery can be performed under local anesthesia by performing a blockade of the superficial and deep cervical plexuses which derive innervation from the C2-C4 nerve roots. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 483.*
Where is the incision for carotid endarterectomy typically made?	The incision is made anterior to the sternocleidomastoid from below the earlobe to the base of the neck. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 406.*
What is the purpose of a shunt during carotid endarterectomy?	If the blood flow from one artery is occluded and collateral circulation is not sufficient, the patient will suffer cerebral ischemia. A shunt alleviates the concerns for insufficient blood flow as it allows blood to bypass the surgical site to the brain while the operation proceeds. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 580.*

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During carotid endarterectomy, a distal stump pressure less than what value is an indicator that a shunt should be placed?	A carotid stump pressure (the pressure in the distal internal carotid artery after carotid clamping) less than 50 mmHg indicates inadequate collateral perfusion which could result in hypoperfusion of the brain during the surgery and is an indicator that a shunt should be placed to ensure adequate perfusion. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 580.*
What are the indications for performing a carotid stent rather than a carotid endarterectomy?	Contralateral laryngeal nerve damage, previous neck radiation therapy, over 80 years of age, severe pulmonary disease, CHF, severe left ventricular function, MI in the last month, unstable angina, and contralateral carotid artery occlusion. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 584.*
How is a carotid stent performed?	The patient receives diagnostic angiograms and/or high-resolution MRI to identify the morphology of the lesion. The femoral artery is cannulated and a sheath is threaded through the femoral artery. It is threaded through the aortic arch and then into the affected carotid artery. A device is deployed which protects the distal artery from emboli during the procedure. Angioplasty of the affected area is performed with a 5 mm balloon and then the stent is deployed. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 584.*
What is the most effective means of assessing cerebral blood flow during a carotid endarterectomy?	A regional anesthetic allows the patient to be awake during the surgery which allows for assessment of the patient's neurologic status during cross-clamping. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 581.*
What is the incidence of significant coronary artery disease in the population of patients undergoing carotid endarterectomy?	30% of patients undergoing carotid endarterectomy have significant CAD. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 407.*
What are the advantages to using a general anesthetic technique for carotid endarterectomy?	It provides a motionless field, overcomes the need for patient cooperation during the surgery, and the use of volatile agents provide a degree of protection against cerebral ischemia by decreasing the CMRO2. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 582.*
What are the advantages to performing an 'awake' carotid endarterectomy with regional anesthesia?	Studies have demonstrated that CEA performed under regional anesthesia is associated with a lower risk for perioperative stroke, a 10% decrease in the risk of perioperative MI, fewer adverse cardiac events, and less intraoperative hemodynamic variability. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1020.*

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What type of regional anesthetic is performed for an awake carotid endarterectomy?	A deep and superficial cervical plexus block with supplementation by a local field block as needed. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 137.*
What is the purpose of performing a carotid endarterectomy under regional anesthesia?	It allows the surgeon to perform a 'test occlusion' of the carotid artery to determine if collateral circulation to the brain is adequate to prevent cerebral ischemia. If after 2-3 minutes, the patient did not experience neurologic changes, the surgery proceeds normally. If the patient develops symptoms, a shunt is inserted or the procedure is converted to a general anesthetic. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 137.*
What is the main disadvantage of an 'awake' carotid endarterectomy with regional anesthesia?	If the patient suddenly develops neurologic symptoms 10 minutes into the procedure or experiences a seizure, general anesthesia must be instituted and the conditions for conversion to a general anesthetic are not ideal this late into the procedure. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 137.*
How does postoperative hypertension following carotid endarterectomy correlate with morbidity?	Patients who exhibit a systolic blood pressure greater than 180 mmHg exhibit an increased incidence of stroke, TIA, and myocardial infarction. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 582.*
How should ventilation be managed during a carotid endarterectomy?	You should strive to maintain normocarbia. Both hypercarbia and hypocarbia can result in alterations in cerebral blood flow. Hypocarbia can constrict cerebral blood vessels and potentially reduce blood flow. Hypercarbia can dilate vessels and potentially result in a steal phenomenon in which areas of decreased perfusion suffer increased ischemia at the expense of increased perfusion to normal vessels. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1021.*
Is anticoagulation utilized during carotid endarterectomy? What about carotid stent placement?	Yes, heparin in the range of 5,000-10,000 units are administered for both procedures. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 585.*
Which ECG leads would be most appropriate to monitor in a patient undergoing carotid endarterectomy?	Patients undergoing carotid endarterectomy are at high risk for ischemic events during the perioperative period. Monitoring leads II and V5 offers the greatest opportunity for detecting ischemic changes should they occur. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1126.*

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What is the dose of protamine for the reversal of heparin following a carotid procedure?	Protamine is administered with a dose of 0.5 mg per 100 units of heparin given about 10 minutes after the repair is complete. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 137.*
What nerves are at risk for damage during a carotid endarterectomy?	Phrenic, vagus, recurrent and superior laryngeal nerves, ansa hypoglossi, and hypoglossal *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 406.*
What drugs should be used to control blood pressure during a carotid endarterectomy and why?	During a carotid endarterectomy, it is advised to maintain the patient's MAP slightly above their highest recorded preoperative pressure (usually a MAP between 80-100 mmHg). For severe hypertension, labetalol may be used, but nitroglycerin is preferred for its swift onset and short duration of action as labetalol could result in postoperative hypotension due to its long duration of action. For hypotension, a pure alpha-agonist such as phenylephrine is preferred because it has minimal dysrhythmogenic potential. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 139.*
Following a carotid endarterectomy, the surgeon asks the patient to smile, say "EEE", shrug his shoulders, swallow, and stick his tongue out. What specific cranial nerve is each of these tests assessing?	Smile: facial nerve, say "EEE": superior and recurrent laryngeal nerves, shrug shoulders: spinal accessory nerve, swallow: glossopharyngeal, stick his tongue out: hypoglossal nerve. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 583.*
During carotid endarterectomy, a patient exhibits profound bradycardia. What is the likely cause of this, how should it be treated, and how can a recurrence be prevented?	Bradycardia caused by manipulation of the baroreceptors in the carotid sinus should initially be treated with atropine. To lessen the likelihood that this will occur again, the surgeon can infiltrate the area around the carotid sinus with lidocaine. Beware however, that the infiltration itself can precipitate a baroreceptor response. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 482.*
Why is it important to avoid hypotension during cross-clamping of the carotid artery?	During carotid cross-clamping, it is imperative that the patient suffer no intraoperative hypotension because the areas of the brain served by the clamped carotid artery will rely on blood flow from the contralateral carotid artery through the circle of Willis. For this reason, it is important to maintain adequate perfusion pressure so that the side that is clamped is not hypoperfused. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 135.*

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What is the mortality rate following a carotid endarterectomy?	0.5-2.5% *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 579.*
What is the relationship between the occurrence of a transient ischemic attack and the occurrence of cerebrovascular accidents?	Over half of all strokes are preceded by a transient ischemic attack. The risk of stroke is 30% within two years of the first TIA. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 578.*
What is the perioperative incidence for myocardial infarction when a carotid endarterectomy is performed?	2-5%. Myocardial infarction is the greatest contributor to the risk of morbidity following a carotid endarterectomy. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 579.*
What is the incidence of males undergoing carotid endarterectomy compared to that of females?	Males are three times more likely to undergo a carotid endarterectomy than females. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 407.*
What is the incidence of damage to the recurrent and superior laryngeal nerves during carotid endarterectomy?	39% *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 407.*
What is a pharmacologic method for decreasing the cerebral metabolism for protection against ischemia during periods of decreased cerebral blood flow?	Propofol has been shown to reduce cerebral metabolism by 40%. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 580.*
Unilateral blindness occurs in 25% of patients presenting for carotid endarterectomy. What is the pathophysiology behind this symptom?	Microthrombi break free from the carotid lesion and travel into the internal carotid artery and then obstruct the opthalmic artery. *Nagelhout JJ, Zaglaniczny KL. Nurse Anesthesia. 4th ed. Philadelphia, PA: WB Saunders Company; 2010: 547.*
What is the incidence of correctable coronary artery disease in patients presenting for carotid endarterectomy and what does this imply?	26% of patients undergoing carotid endarterectomy have correctable coronary artery disease. This implies that close preoperative evaluation of patients presenting for this surgery may identify those that need further cardiac assessment and possibly treatment prior to undergoing surgery. Not all patients will need a cardiac catheterization with stenting, but approximately 1/4 of them may require some therapeutic intervention which would dramatically decrease their risk for perioperative morbidity. *Nagelhout JJ, Zaglaniczny KL. Nurse Anesthesia. 4th ed. Philadelphia, PA: WB Saunders Company; 2010: 547.*

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What are the three major branches of the common carotid artery near the bifurcation?	The internal carotid artery, the external carotid artery, and the superior thyroid artery. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 134.*
What methods of monitoring cerebral perfusion are commonly used during carotid cross-clamping during a carotid endarterectomy?	EEG, somatosensory evoked potentials, internal carotid artery stump pressure, and transcranial doppler. During awake carotid endarterectomy, the patient's neurologic status and responsiveness is the indicator of cerebral perfusion. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 135.*
How should blood pressure be maintained during carotid cross-clamp?	Mild hypertension (a MAP between 90 and 110) should be maintained during cross-clamping. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 135.*
How should blood pressure be maintained at the end of a carotid endarterectomy?	You should strive to maintain a mean arterial pressure between 80 and 100 mmHg. Hypotension could result in loss of patency of the carotid artery while hypertension could stress the fragile tissues that were underneath the lesion or the incision site in the carotid artery and result in massive hemorrhage. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 135.*
What is the most common cause of occlusive vascular disease of the lower extremities?	Atherosclerosis *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 561.*
What are the symptoms of peripheral vascular disease?	Claudication, skin ulcers, gangrene, and impotence. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 561.*
What is the incidence of coronary artery disease in patients presenting for vascular surgery?	The incidence of significant coronary artery disease (stenosis >70%) in patients presenting for vascular surgery is 37% in patients with no symptoms of CAD and 78% in patients with symptoms of CAD. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1115.*
What are the two major contributing factors to the development of peripheral vascular disease?	Diabetes mellitus and smoking *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 561.*

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What types of surgeries may be implemented to treat ischemia from peripheral vascular disease?	Transluminal angioplasty, endarterectomy, thrombectomy, stenting, and arterial bypass surgery. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 561.*
What is the most common causative factor in the mortality of patients undergoing surgery for vaso-occlusive disease?	Events related to atherosclerotic cardiovascular disease are responsible for more than half of the perioperative deaths in patients undergoing surgery for vaso-occlusive disease. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 561.*
What are the cardiovascular benefits of using epidural anesthesia technique for patients undergoing vascular surgery?	It decreases myocardial afterload, MVO2, is associated with less variability in sympathetic tone, decreased endocardial ischemia, results in less blood loss, and redistributes blood to the lower extremities. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 562.*
What are the renal benefits of using epidural anesthesia technique for patients undergoing vascular surgery?	It results in less renovascular constriction and increases blood flow to the renal cortex. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 562.*
What are the endocrine benefits of using epidural anesthesia technique for patients undergoing vascular surgery?	It inhibits the stress response to surgery and thereby inhibits catecholamine and cortisol release. Epidural anesthesia also inhibits hyperglycemia, lymphopenia, and granulocytosis, and blocks sympathetic tone which aids in revascularization of the lower extremities. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 562.*
What vessels comprise the arterial supply to the spinal cord?	Two posterior spinal arteries provide 25% of spinal cord blood flow. The anterior spinal artery, which arises from a series of aortic radicular arteries comprises the other 75% of the arterial flow. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1122.*
What are the pulmonary benefits of using an epidural anesthesia technique for patients undergoing vascular surgery?	It is associated with decreased V:Q mismatching, improves oxygenation, decreased incidence of thromboembolism, and decreased incidence of postoperative pulmonary complications. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 562.*

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Why is hypotension such a significant factor in spinal cord perfusion?	The flow to the spinal arteries depends on collateral supply. Periods of hypotension will shunt a significant portion of the spinal cord blood supply to the mesentery and can result in spinal cord ischemia. This scenario can occur with the placement of a single, high aortic cross-clamp. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1122.*
What is the incidence of spinal cord ischemia in patients undergoing surgery involving aortic cross-clamp?	The incidence is between 1% and 11% *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1122.*
What is the primary blood supply to the thoracolumbar spinal cord?	The artery of Adamkiewicz is an aortic radicular artery that supplies most of the blood flow to the anterior spinal artery. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1122.*
At what vertebral level does the artery of Adamkiewicz connect with the anterior spinal artery?	In 75% of patients, it joins the anterior spinal artery between T8 and T12. In the other 10%, it connects between L1 and L2. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1122.*
What are the risk factors for abdominal aortic aneurysm formation?	Advanced age, smoking more than 40 years, hypertension, low HDL cholesterol, high plasma fibrinogen levels, and low platelet count. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1128.*
Aortic aneurysms can be large enough to compress other anatomic structures. What structures are commonly compressed and what symptoms are produced?	Patients with thoracic aortic aneurysm are often asymptomatic initially. Early presenting symptoms are due to compression of anatomic structures by the aneurysm. Compression of the esophagus can result in dysphagia. Compression of the trachea can cause stridor. Compression of the left recurrent laryngeal nerve can cause hoarseness. Compression of the lungs can cause dyspnea. Compression of the vena cava can cause edema. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 155.*
How does the incidence of abdominal aortic aneurysm formation differ between males and females?	The incidence is four times greater in male patients. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1128.*

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What is the mortality rate for surgical repair for an intact abdominal aortic aneurysm?	The rate ranges (depending on the study) between 1% and 11%, but most resources report a 5% mortality. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 5596.*
What is the mortality rate from a ruptured aortic aneurysm?	85% *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1128.*
What is the factor that determines the risk of rupture of an abdominal aortic aneurysm?	The absolute diameter of the aneurysmal sac is the primary determinant of the risk of rupture. The risk for rupture over a 6 year period is 1% for an aneurysm less than 4 cm in diameter, 2% for aneurysms between 4.0 and 4.9 cm in diameter, and 20% for aneurysms greater than 5 cm. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1128.*
How does blood pressure change above and below the clamp when the aorta is cross-clamped?	Hypertension exists above the level of the clamp and hypotension occurs below the level of the clamp. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 564.*
How are the mean arterial pressure and systemic vascular resistance affected by the release of an aortic cross-clamp?	They both decrease *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 566.*
How does aortic cross-clamping affect mean arterial pressure and systemic vascular resistance proximal to the clamp?	Aortic cross-clamping increases the MAP and SVR by about 50%. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1134.*
What is the most sensitive indicator of cardiac ischemia during thoracic surgery when the aorta is cross-clamped?	Although TEE and 12-lead ECG have little advantage over preoperative clinical data and two-lead ECG monitoring in non-cardiac surgery, the advantage of TEE during cardiac or thoracic surgery is clear. In one study, TEE detected wall-motion abnormalities in 11 out of 12 patients, while the pulmonary artery occlusive pressure remained unchanged in 10 of them. Additionally, during the period of ischemia, no significant changes occurred in the heart rate or systolic pressure. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 129-130.*

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What pharmacologic agent has been shown to decrease the morbidity rate 10-fold in patients undergoing surgery for aortic aneurysm repair and what is the mechanism for its action?	Because these patients have a high incidence for coronary artery disease, the ability of beta-blockers to improve the balance between supply and demand in the myocardium has been shown to reduce the perioperative morbidity of patients undergoing aortic aneurysm repair ten-fold. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 561.*
What commonly used techniques to preserve renal function during aortic clamping have shown to be of little benefit?	The administration of mannitol, dopamine, and fenoldapam have all been shown to have little consequence on the development of renal failure following aortic clamping. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1120.*
How does the incidence of acute renal failure following abdominal aortic aneurysm repair compare when the surgery involves infrarenal aortic clamping versus suprarenal aortic clamping?	Infrarenal clamping is associated with a lower incidence of postoperative acute renal failure than suprarenal clamping. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1120.*
What are the preventative measures that reduce the risk of spinal cord ischemia during aortic cross-clamping?	Short surgery, short aortic cross-clamp duration, preservation of normal cardiac function, and high perfusion pressures. Hypothermia and CSF drainage can also reduce the risk of spinal cord ischemia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1133.*
How can you offset the drop in mean arterial pressure and SVR associated with the release of an aortic cross-clamp?	In anticipation of the ensuing hypotension, it is prudent to administer IV fluids to normalize the preload, discontinue any vasodilators currently infusing, and administer sodium bicarbonate to offset the acidosis (this is common practice, but is controversial with some researchers speculating that it increases the risk of hypotension by increasing acidity within cardiac myocytes). *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 421.*
How is mannitol beneficial when administered before and after the release of an aortic cross-clamp?	Mannitol increases urine output during aortic cross-clamp, but this effect has not been shown to alter renal outcomes after surgery involving aortic cross-clamping. It does, however serve as a free radical scavenger which may be of benefit after the release of the aortic cross-clamp. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1120.*

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How is the cardiac output affected by the release of an aortic cross-clamp?	It decreases slightly. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 566.*
How is the pulmonary artery pressure affected by the release of an aortic clamp?	It increases *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 566.*
How does the presence of preoperative renal insufficiency affect the outcome of patients undergoing abdominal aneurysm repair?	The most important predictor of acute renal failure after thoracic aortic surgery is preoperative renal function. The mortality rate is significantly higher in patients with pre-existing renal insufficiency compared to those who undergo AAA surgery with normal renal function. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1120.*
What is typically responsible for most of the blood loss that occurs during an abdominal aortic aneurysm repair?	Most of the blood loss is due to back-bleeding from the lumbar and inferior mesenteric arteries once the vessels have been clamped and the aneurysm incised. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 567.*
What is the most common cause of perioperative mortality following an abdominal aortic aneurysm repair?	Myocardial infarction is responsible for 40-70% of the deaths surrounding abdominal aortic aneurysm repair. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 674.*
Following thoracic aortic surgery, a patient exhibits anterior spinal artery syndrome. What symptoms are associated with this disorder?	Cross-clamping the thoracic aorta introduces a risk for anterior spinal cord syndrome. The incidence is 0.2% for elective infrarenal aortic surgery, 8% for elective thoracic aneurysm repair, and up to 40% for emergency repair of aortic dissection or rupture. The symptoms include flaccid paralysis of the lower extremities and loss of control of the bowels and bladder. Sensation and proprioception are not affected. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 157-158.*
What two techniques have been shown to be effective at preventing acute renal failure following aortic clamping?	Adequate hydration during clamping and after clamp release has been shown to be an important factor in renal outcomes. The direct instillation of crystalloid or blood chilled to 4 degrees Celsius into the renal arteries to produce profound renal hypothermia has also been shown to be protective during periods of protracted ischemia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1120-1121.*

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How does cross-clamping the thoracic aorta affect cerebrospinal fluid pressure?	The cross clamp results in redistribution of the blood volume with vascular engorgement in the cranial vault. The increased pressure forces cerebrospinal fluid into the spinal compartment and increases CSF pressure. A spinal fluid drain is sometimes used to help prevent inordinant increases in CSF pressure. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 158-159.*
How does one-lung ventilation affect intrapulmonary shunting?	During two-lung ventilation, about 60% of the pulmonary blood flow goes to the dependent lung. Therefore, when one-lung ventilation is instituted, you would expect the patient to experience approximately a 40% shunt as blood that is unable to be oxygenated travels through the nondependent lung. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 674.*
What are the side-effects of intentional lung collapse for surgery?	Intentional lung collapse on the operative side is commonly used to enhanced visualization of the operative field. Side effects are primarily due to the large intrapulmonary shunt that develops because the upper lung is still being perfused but not ventilated. The PA-a gradient widens and hypoxemia typically ensues. One-lung ventilation does not appreciably change the arterial CO2 tension. If a patient has a PAC placed in the nondependent lung, and that lung is collapsed for surgical access, the cardiac output and mixed venous oxygen tension may both be falsely decreased. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 549.*
Arterial emboli are most likely to lodge at what location?	Although emboli frequently lodge at the iliac, femoral, or popliteal arteries, 50% of all embolic events involve the common femoral artery. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 438.*
What are the elements of Virchow's triad?	Virchow's triad is a set of three factors that predict a high risk for pulmonary embolism. Virchow's triad includes hypercoagulability, venous stasis, and vessel wall abnormalities. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 629-630.*
What patients are at the highest risk for embolus formation that requires surgical intervention?	Patients with a history of MI, mitral stenosis, or atrial fibrillation. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 438.*

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Vascular Surgery

What artery is typically occluded by upper extremity emboli?	The brachial artery *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 438.*
Where should the arterial line be placed in a patient undergoing surgery on the ascending aorta?	During surgery on the ascending aorta, it may become necessary to clamp the innominate artery which would occlude the right radial artery. Monitoring pressures in the left radial, femoral, or dorsalis pedis arteries would be suitable alternatives. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 478.*
How can hypothermia be used to provide cerebral protection against ischemia?	A decrease in the core temperature by 1 degree Celsius is associated with a 8% decrease in CMRO2. Hypothermia between 12 and 20 degrees Celsius has been shown to protect against ischemia for 30 to 60 minutes. Mild hypothermia (33-35 degrees Celsius) has not been shown to provide significant protection. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 408.*
You are about to anesthetize a patient with end-stage renal disease for the placement of an arteriovenous dialysis graft. What anesthetic medications should you avoid? What medications are considered safe for administration to this patient?	Patients with end-stage renal disease have decreased renal elimination of many drugs, including opioids. Because of the prolonged excretion of the metabolite normeperidine, meperidine should probably be avoided in these patients. Volatile agents, propofol, fentanyl, hydromorphone, remifentanil, nitrous oxide, and sufentanil are generally regarded as acceptable for use in renal failure patients. Single dose morphine is considered acceptable for use in patients with renal failure, but repeated doses or a continuous infusion can result in prolonged sedation. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 664.*
How should you prepare for a patient undergoing an endovascular aortic stent placement?	Patient preparation for an aortic stent is similar to that of an open abdominal aortic aneurysm. Two large bore IVs or a central line and peripheral IV are mandatory. The patient should be typed and screened, if not cross matched. It is also a good idea to have vasoactive drugs hanging and ready to infuse. An arterial line is required in order to assess blood pressure on a beat by beat basis and to fine tune titration of vasopressors. It is important to note that these patients may have significant peripheral vascular disease and it may be extremely difficult to cannulate the radial artery. The anesthetic technique is usually general endotracheal, but the procedure can be performed under an epidural block or even IV sedation. Muscle relaxation is not necessary. *Vascular Web. Endovascular Stents. [Online] June 23, 2008 <http://www.vascularweb.org/patients/NorthPoint/Endovascular_Stent_Graft.html>.*

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What anesthetic plans would be acceptable for a patient undergoing varicose vein stripping?	Anesthesia for varicose vein stripping may be performed under epidural or spinal anesthesia (T10-T12 level usually preferable), local anesthesia using field block (often improved by ilioinguinal and iliohypogastric blocks) with or without sedation, or general anesthesia using ETT or LMA. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 460.*
What pulmonary complications can occur when the thoracic aorta is cross-clamped?	Cross-clamping the thoracic aorta is associated with significant decreases in blood flow in the distal anterior spinal artery and renal artery. Pulmonary complications can arise from aortic cross-clamp due to increased pulmonary vascular resistance with resulting increased pulmonary capillary permeability and pulmonary edema. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 158-159.*
What is a portosystemic shunt used to treat?	Portal hypertension develops when the splanchnic venous return to the heart becomes impeded. This results in congestion of the venous return and leads to esophageal varices, which carries a 50% mortality rate with the first episode of hemorrhage. A portocaval or portosystemic shunt is performed to anastamose the portal vein to the inferior vena cava to decompress the portal venous system. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 445.*

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